Report Case Enter – Rafael Lima Figueiredo

Main issues with the first version of the workflow

- **Prompts are generally too long**, which increases API costs (due to higher token usage) and can lead to recency bias and instruction degradation. - Reference: https://arxiv.org/pdf/2310.01427

- **Lack of guardrails throughout the workflow**. LLMs tend to hallucinate when asked to infer or fill in missing information unless explicitly instructed not to. This is especially concerning when dealing with real portfolio data, as the model may fabricate inputs such as interest rates, inflation values, or investment returns when it lacks direct access to them. - Reference: https://arxiv.org/html/2510.06265v2#S6

- **Non-standardized preparation and reading of input files**. Simply converting PDFs to raw .txt files does not guarantee correct interpretation of the content — especially for macro reports with charts, tables, and mixed formatting (e.g., XP's Macro Analysis).

- **Using the same temperature for all tasks.** For activities such as factual summaries, which require high fidelity to the original content, low linguistic variability, and minimal inference, it is preferable to use lower temperatures to reduce creativity, increase precision, and avoid unwanted assumptions.

- **Using Large Language Models (LLMs) for the automatic filling of fields considered static**, such as the Client Name in documents. This is resulting in hallucinations, where the model generates factually incorrect information; for example, the client's correct name, Albert, is being improperly replaced by John in the final output, compromising communication accuracy.

- **Final letter unformatted, with different fonts and with nonsensical tabs.** This undermines XP's professionalism and credibility.

Rationale behind new approach

The prioritization process was based on the analysis of the three core objectives of the workflow: (i) explaining fluctuations in a client's investment portfolio, (ii) outlining current market trends to clients in lay terms, and (iii) recommending investment opportunities that are aligned with clients' risk profiles.

Among these objectives, explaining monthly portfolio fluctuations was prioritized for three main reasons:

1. It is the most frequent and sensitive client demand

Clients are consistently interested in understanding the performance of the assets within their portfolios. This question is recurring and directly tied to the client's perception of transparency, trust, and service quality. As a result, it typically represents the highest volume request in the advisor's daily routine.

2. It reduces operational burden for the advisor

Advisors manage a large client base and repeatedly addressing questions about monthly performance create significant operational strain. This becomes even more relevant given the volume of interactions required and the need for consistent, accurate answers.

3. It is technically the simplest of the three tasks

Unlike investment recommendations, which require contextual analysis, regulatory caution, and deeper knowledge of the client's profile, performance explanations tend to be more objective and based on historical data. This lowers the risk of misinterpretation and allows for safer automation.

Based on this rationale, a Python script was developed to read the client's portfolio file and extract key metrics, such as the total capital invested and specific holdings across stocks, fixed-income securities, and investment funds.

Subsequently, the workflow utilizes the *yfinance* library to calculate the total monthly return for each stock—accounting for dividends and stock splits—and to generate a visual summary graph of the client's current allocation.

Finally, this aggregated data provides the Large Language Model (LLM) with the necessary context regarding the client's portfolio. This enables the model to offer well-founded opinions on potential investment recommendations, which are automatically incorporated into the final client letter. This approach significantly reduces the advisor's manual effort while delivering enhanced value to the client.

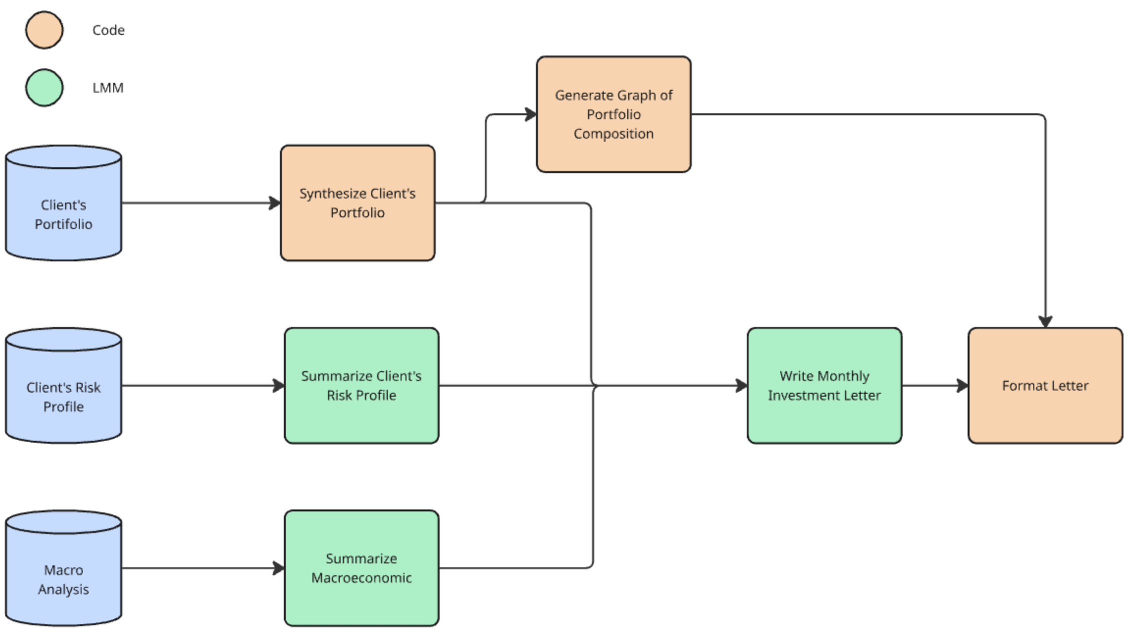

Image representing the proposed workflow.

Future improvements for the project

- **Create a more productive workflow.** It is not necessary to repeat the same summarization process for the macroeconomic report, for example.
- **Stress-test the system with a larger number of clients** — including clients with many assets, clients with only one asset, and clients invested in different asset classes (e.g., ETFs).
- **Improve the macroeconomic scenario parsing**, especially the interpretation of charts and tables, so the model can effectively use this information.
- **Develop a structured portfolio of available investment offerings** across equities, fixed income, and funds, so the model can provide more accurate and profile-aligned recommendations.